

Lecture Slides for

INTRODUCTION TO

Machine Learning

Elmurod Kuriyozov
Department of IT
Urgench State University

CHAPTER 1:

Introduction

Why "Learn" ?

- Machine learning is programming computers to optimize a performance criterion using example data or past experience.
- There is no need to "learn" to calculate payroll
- Learning is used when:
  - Human expertise does not exist (navigating on Mars),
  - Humans are unable to explain their expertise (speech recognition)
  - Solution changes in time (routing on a computer network)
  - Solution needs to be adapted to particular cases (user biometrics)

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

What We Talk About When We Talk About"Learning"

- Learning general models from a data of particular examples
- Data is cheap and abundant (data warehouses, data marts); knowledge is expensive and scarce.
- Example in retail: Customer transactions to consumer behavior:

  People who bought "Da Vinci Code" also bought "The Five People You Meet in Heaven" (www.amazon.com)
- Build a model that is *a good and useful approximation* to the data.

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Data Mining

- Retail: Market basket analysis, Customer relationship management (CRM)
- Finance: Credit scoring, fraud detection
- Manufacturing: Optimization, troubleshooting
- Medicine: Medical diagnosis
- Telecommunications: Quality of service optimization
- Bioinformatics: Motifs, alignment
- Web mining: Search engines
- ...

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

What is Machine Learning?

- Optimize a performance criterion using example data or past experience.
- Role of Statistics: Inference from a sample
- Role of Computer science: Efficient algorithms to
  - Solve the optimization problem
  - Representing and evaluating the model for inference

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Applications

- Association
- Supervised Learning
  - Classification
  - Regression
- Unsupervised Learning
- Reinforcement Learning

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Learning Associations

- Basket analysis:

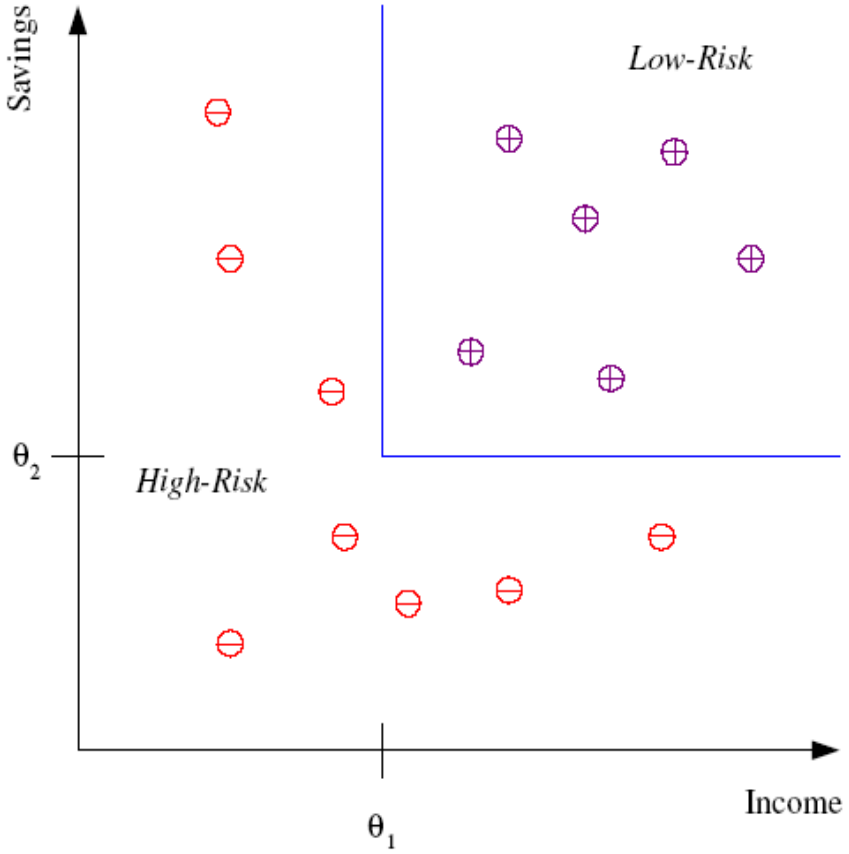
  $P(Y \mid X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

  Example: $P(\text{ chips} \mid \text{beer}) = 0.7$

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Classification

- Example: Credit scoring
- Differentiating between low-risk and high-risk customers from their *income* and *savings*



Discriminant: IF *income* $> \theta_1$ AND *savings* $> \theta_2$
THEN low-risk ELSE high-risk

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Classification: Applications

- Aka Pattern recognition
- Face recognition: Pose, lighting, occlusion (glasses, beard), make-up, hair style
- Character recognition: Different handwriting styles.
- Speech recognition: Temporal dependency.
  - Use of a dictionary or the syntax of the language.
  - Sensor fusion: Combine multiple modalities; eg, visual (lip image) and acoustic for speech
- Medical diagnosis: From symptoms to illnesses
- ...

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Face Recognition

Training examples of a person



Test images



AT&T Laboratories, Cambridge UK
http://www.uk.research.att.com/facedatabase.html

Regression

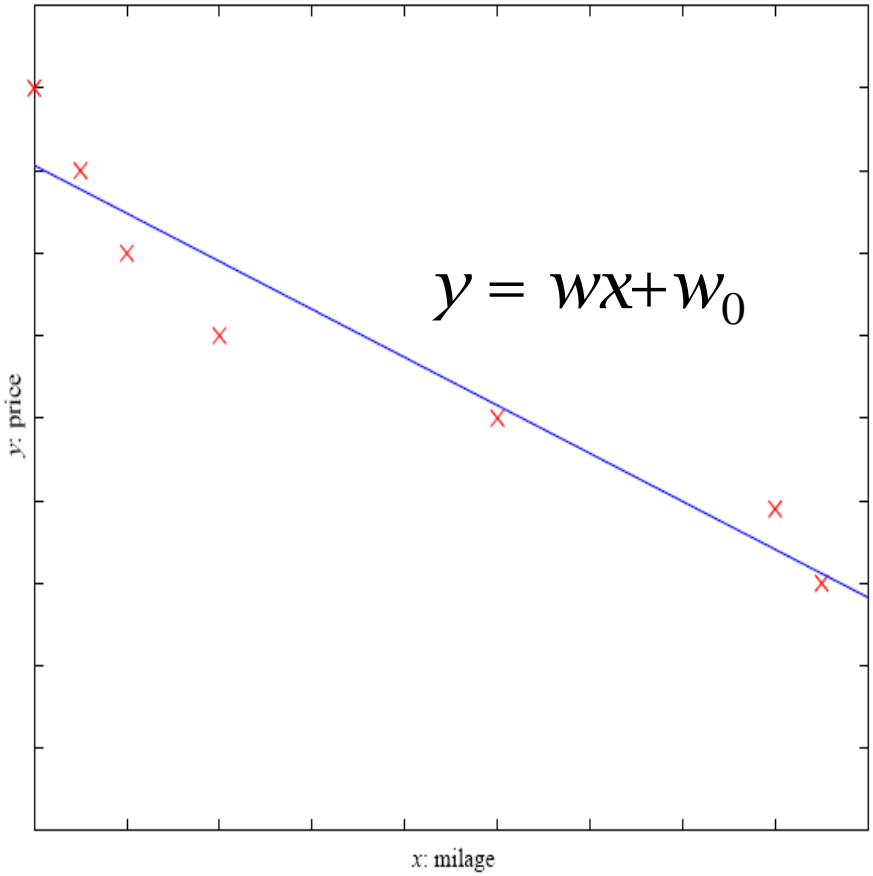
- Example: Price of a used car
- x : car attributes

  y : price

  $$y = g(x \mid \theta)$$

  $g(\ )$ model,

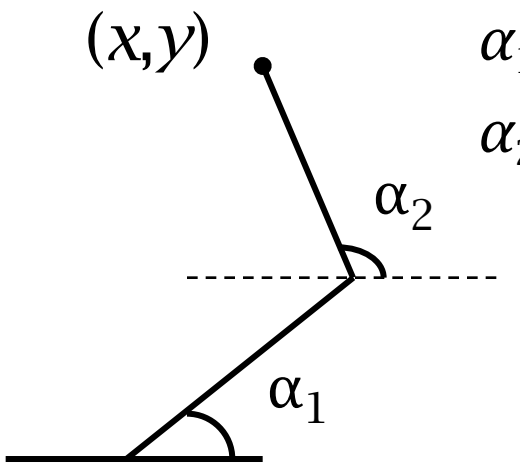
  θ parameters



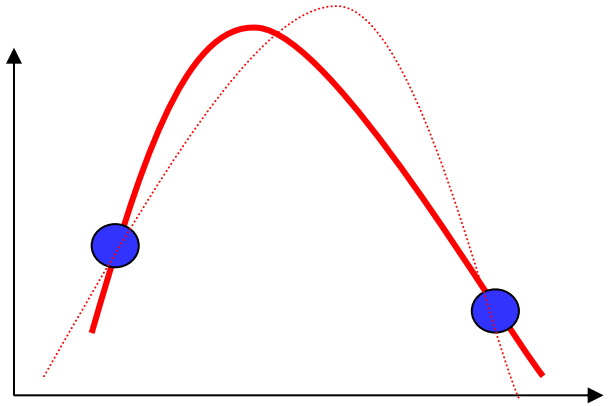
Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Regression Applications

- Navigating a car: Angle of the steering wheel (CMU NavLab)
- Kinematics of a robot arm



$$\alpha_1 = g_1(x,y)$$
$$\alpha_2 = g_2(x,y)$$



- Response surface design

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Supervised Learning: Uses

- Prediction of future cases: Use the rule to predict the output for future inputs
- Knowledge extraction: The rule is easy to understand
- Compression: The rule is simpler than the data it explains
- Outlier detection: Exceptions that are not covered by the rule, e.g., fraud

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Unsupervised Learning

- Learning “what normally happens”
- No output
- Clustering: Grouping similar instances
- Example applications
  - Customer segmentation in CRM
  - Image compression: Color quantization
  - Bioinformatics: Learning motifs

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Reinforcement Learning

- Learning a policy: A sequence of outputs
- No supervised output but delayed reward
- Credit assignment problem
- Game playing
- Robot in a maze
- Multiple agents, partial observability, ...

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Resources: Datasets

- UCI Repository: http://www.ics.uci.edu/~mlearn/MLRepository.html
- UCI KDD Archive: http://kdd.ics.uci.edu/summary.data.application.html
- Statlib: http://lib.stat.cmu.edu/
- Delve: http://www.cs.utoronto.ca/~delve/

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Resources: Journals

- Journal of Machine Learning Research www.jmlr.org
- Machine Learning
- Neural Computation
- Neural Networks
- IEEE Transactions on Neural Networks
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- Annals of Statistics
- Journal of the American Statistical Association
- ...

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)

Resources: Conferences

- International Conference on Machine Learning (ICML)
  - ICML05: http://icml.ais.fraunhofer.de/
- European Conference on Machine Learning (ECML)
  - ECML05: http://ecmlpkdd05.liacc.up.pt/
- Neural Information Processing Systems (NIPS)
  - NIPS05: http://nips.cc/
- Uncertainty in Artificial Intelligence (UAI)
  - UAI05: http://www.cs.toronto.edu/uai2005/
- Computational Learning Theory (COLT)
  - COLT05: http://learningtheory.org/colt2005/
- International Joint Conference on Artificial Intelligence (IJCAI)
  - IJCAI05: http://ijcai05.csd.abdn.ac.uk/
- International Conference on Neural Networks (Europe)
  - ICANN05: http://www.ibspan.waw.pl/ICANN-2005/
- ...

Lecture Notes for E Alpaydın 2004 Introduction to Machine Learning © The MIT Press (V1.1)